

EEG Mathematical Foundations

Part I – Capture, Meaning, and Representation

This note gives the minimum mathematical foundation needed to reason correctly about the EEG signal in our vigilance project. It deliberately stops before filtering, feature extraction, and machine learning.

Mathematical roadmap for the project

Area I – EEG capture, signal meaning, and representation. What quantity does the hardware measure? How is that signal represented, sampled, and compared across acquisition setups?

Area II – Filtering, preprocessing, and feature extraction. How do we decompose the measured signal, suppress unwanted components, and construct the features that represent vigilance?

Area III – Machine-learning model. Given the preprocessed features, how does the model infer vigilance state, and how do we validate that the inference generalizes?

The dependency is

captured signal $\longrightarrow$ preprocessed features $\longrightarrow$ model inference.

This document develops only Area I.

1 What does an EEG channel actually measure?

Conceptual question. An electrode touches the scalp, but what number is stored as an EEG channel?

Conceptual answer. EEG acquisition does not record an absolute potential at one electrode. A channel is a voltage difference between a measurement electrode and a reference electrode. Therefore the reference is part of the measurement itself.

Definition – EEG channel. Let $V_i(t)$ be the electric potential at electrode i and $V_R(t)$ the potential at the reference. The recorded channel is

$$x_i(t) = V_i(t) - V_R(t).$$

Proposition – Common-reference cancellation. If two channels use the same reference, subtracting the channels removes that reference.

Proof. For channels i and j ,

$$x_i(t) - x_j(t) = (V_i(t) - V_R(t)) - (V_j(t) - V_R(t)) = V_i(t) - V_j(t).$$

Thus the common reference cancels exactly. $\square$

Example. If Fz and Cz are both recorded relative to CPz, then

$$x_{Fz} - x_{Cz} = (V_{Fz} - V_{CPz}) - (V_{Cz} - V_{CPz}) = V_{Fz} - V_{Cz}.$$

2 Why can two EEG systems give different signals for the same activity?

Conceptual question. If two datasets both contain a channel called Fz, are those values automatically comparable?

Conceptual answer. No. A channel depends on the reference used to construct it. The same electrode potential can therefore produce different recorded waveforms under different reference schemes.

Suppose two systems use references R_1 and R_2 :

$$x_i^{(1)}(t) = V_i(t) - V_{R_1}(t), \quad x_i^{(2)}(t) = V_i(t) - V_{R_2}(t).$$

Then

$$x_i^{(1)}(t) - x_i^{(2)}(t) = V_{R_2}(t) - V_{R_1}(t).$$

Unless the two reference potentials are identical, the recorded channels are not identical.

Because the Fourier transform is linear, the same statement holds frequency by frequency:

$$X_i(f) = V_i(f) - V_R(f).$$

Thus a reference can change the measured amplitude and power at a given frequency; it does not change what the frequency itself means.

Example. A dataset recording Fz relative to a mastoid measures $V_{Fz} - V_M$. A deployment system recording Fz relative to CPz measures $V_{Fz} - V_{CPz}$. These are different measurements even though both channels are labelled Fz.

Scope note. Scalp electrodes also contain mixtures of underlying neural sources. Recovering those sources would require a separate inverse model. We do not need source localization here because the project uses sensor-level trends and spectral features rather than claims about a specific cortical generator.

3 What is re-referencing, and why can channel number matter?

Conceptual question. If two systems use different references or different numbers of electrodes, can we transform them into a more comparable representation? Does having 12 rather than 32 channels change the mathematical object we observe?

Conceptual answer. Re-referencing constructs new channel differences from already-recorded channels. A common choice is the common-average reference (CAR): subtract the instantaneous average of a chosen channel set from every channel. This removes the original common reference algebraically, but the resulting signal depends on exactly which channels are included. Channel count also determines how many independent sensor coordinates are available: omitted electrodes represent measurements that are simply absent.

Definition – Common-average re-reference. For a chosen set S of N recorded channels, define

$$\bar{x}_S(t) = \frac{1}{N} \sum_{j \in S} x_j(t),$$

and re-reference channel i by

$$x_i^{(S)}(t) = x_i(t) - \bar{x}_S(t).$$

Proposition – The original common reference cancels under CAR. If every channel in S was recorded against the same original reference R , then

$$x_i^{(S)}(t) = V_i(t) - \frac{1}{N} \sum_{j \in S} V_j(t).$$

Proof. Since $x_j(t) = V_j(t) - V_R(t)$,

$$\bar{x}_S(t) = \frac{1}{N} \sum_{j \in S} (V_j(t) - V_R(t)) = \frac{1}{N} \sum_{j \in S} V_j(t) - V_R(t).$$

Therefore

$$x_i^{(S)}(t) = (V_i(t) - V_R(t)) - \left(\frac{1}{N} \sum_{j \in S} V_j(t) - V_R(t) \right) = V_i(t) - \frac{1}{N} \sum_{j \in S} V_j(t).$$

Thus the original reference cancels exactly. $\square$

Proposition – CAR depends on the selected channel set. For two channel sets S and T , the re-referenced value at the same electrode is generally different:

$$x_i^{(S)}(t) - x_i^{(T)}(t) = \frac{1}{|T|} \sum_{j \in T} V_j(t) - \frac{1}{|S|} \sum_{j \in S} V_j(t).$$

Hence $x_i^{(S)} = x_i^{(T)}$ only when the two spatial averages happen to be equal.

Proof. Subtract the two CAR expressions:

$$(V_i - \bar{V}_S) - (V_i - \bar{V}_T) = \bar{V}_T - \bar{V}_S.$$

There is no general reason for two averages taken over different electrode sets to be identical. $\square$

Example. A 32-channel dataset and a 12-channel deployment system can both be called “common-average referenced,” yet

$$\frac{1}{32} \sum_{j=1}^{32} V_j(t) \neq \frac{1}{12} \sum_{j=1}^{12} V_j(t)$$

in general. For cross-system comparison, if the 12 deployment electrodes are available in the 32-channel dataset, first select those same 12 electrodes and then compute CAR over that common set.

Why channel number matters beyond re-referencing. At one time point, an M -channel system records a sensor vector

$$\mathbf{x}(t) = (x_1(t), \dots, x_M(t))^T \in \mathbb{R}^M.$$

Keeping only a subset of $N < M$ channels is a coordinate projection $P : \mathbb{R}^M \rightarrow \mathbb{R}^N$. In general, P is not injective: two full sensor vectors can agree on the retained electrodes while differing on omitted electrodes. Therefore the omitted measurements cannot be recovered from the smaller montage without additional assumptions. More channels do not automatically make a model better, but they provide additional spatial measurements and potential features.

Proposition – Fixed reference: channel count does not change the amplitude of an existing channel. Suppose electrode i is recorded against the same fixed reference R in two systems, and no later re-referencing is applied. Then adding or removing other recorded channels does not change

$$x_i(t) = V_i(t) - V_R(t).$$

Consequently, its Fourier coefficient $X_i(f)$, Fourier amplitude $|X_i(f)|$, and power $|X_i(f)|^2$ are unchanged in the idealized case. Channel count changes which additional amplitude measurements are available, not the amplitude of an already-existing channel.

Proof. The expression $x_i(t) = V_i(t) - V_R(t)$ contains neither the number of recorded channels nor the potentials of any other electrodes. Therefore changing the recorded set while holding V_i and V_R fixed leaves $x_i(t)$ unchanged. By linearity and uniqueness of the Fourier transform, identical time-domain signals have identical Fourier coefficients, so

$$X_i^{(1)}(f) = X_i^{(2)}(f), \quad |X_i^{(1)}(f)| = |X_i^{(2)}(f)|.$$

Thus channel count alone does not alter the amplitude information of a retained channel under a fixed reference. $\square$

Proposition – CAR: channel set can change amplitude information. Under common-average re-referencing over a channel set S ,

$$X_i^{(S)}(f) = X_i(f) - \frac{1}{|S|} \sum_{j \in S} X_j(f).$$

Therefore changing S generally changes the Fourier coefficient and hence the measured amplitude $|X_i^{(S)}(f)|$ and power $|X_i^{(S)}(f)|^2$ at electrode i .

Proof. From

$$x_i^{(S)}(t) = x_i(t) - \frac{1}{|S|} \sum_{j \in S} x_j(t)$$

and linearity of the Fourier transform,

$$X_i^{(S)}(f) = X_i(f) - \frac{1}{|S|} \sum_{j \in S} X_j(f).$$

For two channel sets S and T , these coefficients differ by the difference between the two channel-average spectra. Unless those averages are equal at frequency f , the coefficients, and therefore their magnitudes and powers, differ. $\square$

Example. With the same fixed reference, Fz recorded in a 12-channel system and Fz recorded in a 32-channel system have the same idealized waveform and spectral amplitude if Fz and the reference are unchanged. The 32-channel system simply contains 20 additional spatial measurements. If both systems are instead common-average re-referenced over all available channels, the 12-channel and 32-channel averages are generally different, so the re-referenced Fz amplitude can also differ.

Example. If a 32-channel dataset contains a vigilance-related feature at an electrode absent from the 12-channel system, a model that uses that feature cannot be deployed unchanged. The safe comparison is to train and evaluate using the electrode subset actually available at deployment.

4 How do we represent the measured EEG signal?

Conceptual question. Once a voltage difference is recorded over time, what mathematical object are we working with?

Conceptual answer. Before digitization, we model the measured channel as a continuous function of time, $x(t)$. After sampling, the computer stores a finite sequence of values, which is naturally a vector.

Definition – Continuous-time signal. A channel is modeled as a function

$$x : \mathbb{R} \rightarrow \mathbb{R}, \quad t \mapsto x(t).$$

The value $x(t)$ is the measured voltage difference at time t .

Definition – Uniform sampling. For sampling period $T_s > 0$, define

$$x[n] = x(nT_s), \quad n \in \mathbb{Z}.$$

The sampling frequency is

$$f_s = \frac{1}{T_s}.$$

A finite window of N samples is the vector

$$\mathbf{x} = (x[0], x[1], \dots, x[N-1])^\top.$$

Example. At $f_s = 500$ Hz, $T_s = 1/500 = 0.002$ s. The system records one sample every 2 ms.

5 Which continuous frequencies can discrete samples distinguish?

Conceptual question. Since the computer only sees discrete sample points, can two different continuous oscillations produce exactly the same stored sequence?

Conceptual answer. Yes. Sampling makes continuous frequencies separated by integer multiples of the sampling rate indistinguishable. Frequencies above half the sampling rate therefore fold onto lower frequencies: aliasing.

Proposition – Frequency equivalence under sampling. For the complex sinusoid $x_f(t) = e^{i2\pi ft}$ sampled at rate f_s ,

$$x_f[n] = e^{i2\pi fn/f_s}.$$

For every integer k ,

$$x_{f+kf_s}[n] = x_f[n].$$

Proof.

$$x_{f+kf_s}[n] = e^{i2\pi(f+kf_s)n/f_s} = e^{i2\pi fn/f_s} e^{i2\pi kn} = e^{i2\pi fn/f_s},$$

because $e^{i2\pi kn} = 1$ for integers k, n . $\square$

This means discrete frequency is periodic with period f_s . A unique representative interval is therefore

$$[-f_s/2, f_s/2).$$

The largest magnitude in this interval is the **Nyquist frequency**

$$f_N = f_s/2.$$

Example. Let $f_s = 100$ Hz. A 60 Hz cosine and a 40 Hz cosine give the same samples:

$$\cos(2\pi(60)n/100) = \cos(2\pi n - 2\pi(40)n/100) = \cos(2\pi(40)n/100).$$

So 60 Hz aliases to 40 Hz.

Proposition – Nyquist–Shannon condition. If the continuous signal is band-limited to $|f| < B$, then non-overlapping spectral copies require

$$f_s > 2B.$$

Proof. Sampling replicates the continuous spectrum every f_s Hz:

$$X_s(f) = f_s \sum_{k \in \mathbb{Z}} X(f - kf_s).$$

The central copy occupies $[-B, B]$; the neighboring copy centered at f_s begins at $f_s - B$. To prevent overlap,

$$B < f_s - B,$$

so $f_s > 2B$. When the copies overlap, multiple continuous-frequency components contribute to the same sampled frequency, so the original spectrum is no longer uniquely recoverable. $\square$

6 How can the same sampled signal be described by frequencies?

Conceptual question. The EEG is stored as values over time. How can we rigorously say that it contains a certain amount of 10 Hz, 20 Hz, and so on?

Conceptual answer. A sampled EEG window is a vector. The discrete Fourier transform (DFT) expresses that same vector in another orthonormal basis whose directions are sampled oscillations. Fourier coefficients are therefore coordinates along frequency directions.

Definition – Fourier basis. For N samples, define

$$\phi_k[n] = \frac{1}{\sqrt{N}} e^{i2\pi kn/N}, \quad k, n = 0, \dots, N-1.$$

Each ϕ_k is a sampled complex sinusoid.

Proposition – The Fourier vectors are orthonormal.

$$\langle \phi_k, \phi_\ell \rangle = \delta_{k\ell}.$$

Proof. Using the complex inner product,

$$\langle \phi_k, \phi_\ell \rangle = \frac{1}{N} \sum_{n=0}^{N-1} e^{i2\pi(\ell-k)n/N}.$$

If $k = \ell$, the sum is N , so the inner product is 1. If $k \neq \ell$, set $r = e^{i2\pi(\ell-k)/N}$. Then

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r} = 0,$$

because $r^N = e^{i2\pi(\ell-k)} = 1$ while $r \neq 1$. Hence distinct Fourier vectors are orthogonal. Since there are N orthonormal vectors in $\mathbb{C}^N$, they form a basis. $\square$

Definition – Unitary DFT coefficient. The coordinate of $\mathbf{x}$ along ϕ_k is

$$X[k] = \langle \phi_k, \mathbf{x} \rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-i2\pi kn/N}.$$

Because the Fourier vectors form a basis,

$$\mathbf{x} = \sum_{k=0}^{N-1} X[k] \phi_k.$$

Thus $X[k]$ is literally the coefficient required to reconstruct the measured signal along frequency direction k .

Example. If a signal is composed mainly of a 10 Hz oscillation plus a smaller 20 Hz oscillation, the magnitude of the Fourier coefficient at 10 Hz will be larger. This means more of the measured signal vector lies along the 10 Hz basis direction.

7 Why does squared Fourier magnitude correspond to power?

Conceptual question. Why do spectral methods use $|X[k]|^2$ rather than just $X[k]$?

Conceptual answer. The Fourier coefficient is generally complex: it carries both magnitude and phase. Squared magnitude removes phase/sign dependence and measures the squared size of the component. Parseval's theorem shows that these squared frequency components exactly partition the squared magnitude of the original sampled signal, under unitary normalization.

Proposition – Parseval's identity for the unitary DFT.

$$\sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |X[k]|^2.$$

Proof. Since $\mathbf{x} = \sum_k X[k] \phi_k$,

$$\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle = \sum_k \sum_\ell \overline{X[k]} X[\ell] \langle \phi_k, \phi_\ell \rangle.$$

Orthonormality gives $\langle \phi_k, \phi_\ell \rangle = \delta_{k\ell}$, so every cross-term vanishes:

$$\|\mathbf{x}\|^2 = \sum_k |X[k]|^2.$$

But $\|\mathbf{x}\|^2 = \sum_n |x[n]|^2$, proving the identity. $\square$

Therefore $|X[k]|^2$ is the squared-magnitude contribution associated with frequency bin k .

For a set of frequency bins A , a band quantity can be defined as

$$P_A = \sum_{k \in A} |X[k]|^2.$$

For example, an alpha-band quantity sums the relevant bins rather than treating a single frequency in isolation.

Example. For a pure sinusoid $x(t) = A \sin(\omega t)$ over whole cycles,

$$\frac{1}{T} \int_0^T x(t)^2 dt = \frac{A^2}{2}.$$

Thus the mean-square size of the oscillation scales with amplitude squared, consistent with the squared-magnitude interpretation in the frequency domain.

8 What should we carry into the next stage?

The key mathematical object delivered to preprocessing is a multichannel sampled voltage signal. For channel i ,

$$x_i[n] = V_i(nT_s) - V_R(nT_s).$$

The important consequences are:

- channels are reference-dependent voltage differences;
- two acquisition systems are not automatically 1:1 comparable even when electrode names match;
- re-referencing can remove a shared original reference, but the result depends on the selected channel set;
- with a fixed reference, channel count changes which amplitude/spatial features are available but does not by itself change the amplitude of a retained channel; under CAR, the selected channel set can also change that channels waveform, Fourier amplitude, and power;
- sampling converts a continuous signal into a finite-dimensional vector;
- frequencies above $f_s/2$ are not uniquely represented and can alias;
- the DFT is an orthonormal change of basis from sample coordinates to frequency coordinates;
- $X[k]$ is the coordinate along a frequency direction;
- $|X[k]|^2$ partitions the signal's squared magnitude by frequency via Parseval;
- we stay in sensor space: we describe trends in recorded channels without claiming a unique cortical source.

This closes Area I. Area II begins only when we decide which vigilance features matter and therefore what filtering, artifact handling, normalization, and feature extraction are required.